

User guide.

A comprehensive guide to using your DiscoX survey device, if anythings missing let me know and I'll get it updated.

Showing the original device – four identical blue buttons, no orange trigger.

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Buttons

The DiscoX has four buttons.



In normal mode

- **Button 1 (Top)** — take a measurement, or wake the laser if it's timed out.
- **Button 2** — short press: take a measurement. Hold 2s: toggle Disco mode.
- **Button 3** — open the settings menu.
- **Button 4 (Bottom)** — power on/off.

In **menu mode**, Buttons 1/2/3 become scroll-up, scroll-down, and select. Button 4 still powers the device off.

Taking readings

Press **Button 1** or **Button 2** (short press) to take a reading. The device waits until the sensors have settled before accepting a shot — hold the device still during readings.

LED states



RED

Stabilising — hold the device steady.



GREEN

Stable — shot recorded and sent.



PURPLE

Leg detected — three beeps confirm.

After a shot the display freezes on the recorded reading and the laser turns off. Press any measure button to re-enable the laser, then press again to take the next reading.

Leg detection

When three consecutive shots agree in bearing, clino, and distance (within configurable tolerances), the DiscoX recognises a complete survey leg. The LED pulses purple, the laser flashes briefly, and the buzzer beeps three times. The display stays frozen on the leg data until you press a measure button to continue.

Laser wake

If the laser has auto-timed-out, pressing any measure button re-enables it with a short beep before taking the shot.

Display

The OLED display shows three values from your most recent stable shot:

- **Distance** (top) — in metres, to two decimal places.
- **Azimuth** (middle) — compass bearing, in degrees.
- **Inclination** (bottom) — clino angle, in degrees.

The **top-left** shows  when Bluetooth is connected, with a count of pending (unsent) readings beside it. The **top-right** shows a proportional battery bar. If the device is set to measure from the front this is indicated at the top of the display as well.

Disco mode

Hold **Button 2 for 2 seconds** to toggle disco/rave mode. The laser shuts off and the NeoPixel cycles through colours. **Shake the device** for faster colour cycling and full brightness.

To exit press any of the buttons other than the power button. The laser restarts and the device returns to normal operation.

Bluetooth & memory

The DiscoX works with current versions of **SexyTopo** and **TopoDroid** via a long-range C++ implementation of the SAP6 BLE protocol. The device advertises continuously, so nearby phones and tablets can discover it, unpair and repair it from the Bluetooth settings on your phone.

Once connected, the device flashes blue, a **BT** label appears in the top-left of the screen, and live readings are pushed directly to your survey app.



NOTE • BLE NAMING

Each DiscoX ships with a unique Bluetooth name, this can be changed from within the **settings.json** config file by changing the **ble_name** config item. Change it via the settings file, or request a specific name before assembly. Note the name has a limit of 20 characters.

Calibration

Calibrate whenever you survey in a new area or if readings seem off. Run a **Mag Field Check** before each trip to quickly verify accuracy at your survey site.

To start: press **Button 3** → **Enter Calibration**.

Calibration is performed in two stages. **Stage 1** corrects for sensor bias; **Stage 2** corrects for the alignment between the sensors and the laser. Together they take 80 readings.

TAKE IT UNDERGROUND

PRINTABLE CALIBRATION CARD

Both stages, the field check and the on-screen warnings on one sheet. Print A4 landscape at 100%, double-sided, and fold down the middle for a pocket-sized A5 card.

[Download PDF](#)

Stage 1 — Sensor bias

With perfect sensors under perfect conditions, taking survey readings in every orientation would trace out a sphere. In practice, manufacturing variance between sensors and the presence of metal components mean an uncalibrated device traces something closer to a rugby ball.

The fix is to take readings across a well-distributed set of orientations, then compute a correction matrix that can be applied to the raw sensor readings to give a less biased output.

56 READINGS

Taking the readings

You could take readings in random directions and still get a decent result, but a recommended set usually does better. For the DiscoX, aim roughly at every face and corner of an imaginary cube — 14 positions — rolling the device 90° four times at each one.

This is similar to the DistoX, but with no need to worry about where exactly the laser is pointing.

Do try to keep the device still as you take each reading. Rest your arms on a non-magnetic surface while moving it, or use a jig — this helps ensure the accelerometer readings are correct.

B1 record • B3 undo
B1+B2 save • B2 hold discard

Stage 2 — Sensor alignment

When the sensors are mounted in the device it is very difficult to get them perfectly aligned with the laser, so an alignment function is needed to correct for the residual error.

24 READINGS

Taking the readings

Point the device at a target and barrel-roll it, taking **8 readings** evenly distributed around the same axis as the laser. Repeat this twice more, aiming at targets in completely different directions.

B1 record • B3 undo
B1+B2 save • B2 hold discard

! HOLD IT STEADY

For this stage it is very important that the readings are on point and the device is steady. The best way I have found is to sit down with your head and back supported, rest the back of the device against your nose, then lock your arms to your sides and take the readings as described above. This gives the most stable base, because there are very few degrees of freedom between you and the device.

Mag field check

Verifies calibration accuracy without re-running a full calibration.

FIELD CHECK

How it works

Shoot a fixed target, walk to it, shoot back. Repeat for **3+ pairs** in different directions. The device records each pair's bearing error and calculates an overall correction amplitude.

A consistently high amplitude means it's time to run a full calibration.

- B1 take shot
- B2 finish (after 2+ pairs)

Quality metrics

After calibration the screen shows three quality scores. **Lower is better** in all cases. View them any time via **Menu** → **Enter Calibration** → **View Last Cal.**

- **Mag** — magnetometer ellipsoid fit uniformity.
- **Grav** — accelerometer ellipsoid fit uniformity.
- **Acc** — final accuracy in degrees (the most important number).

EXCELLENT < 0.25°  ACCEPTABLE < 0.5°

TIP

For best results, calibrate in the same magnetic environment as your survey — i.e. away from bolts, helmet lights, and SRT kit. Anomaly detection will warn you if your calibration is being contaminated.

Firmware update

Go to **Menu** → **Update / Settings** → **Update Firmware**. The device enters the UF2 bootloader and appears as a removable drive. Drag and drop the new  file onto it — the DiscoX reboots when flashing is complete.

Settings

All common settings are accessible from the onboard menu — press **Button 3** to open it.

Onboard menu

- **Enter Calibration** — run a calibration, perform a Mag Field Check, or view last results.
- **Anomaly Detection** — toggle warnings for nearby metallic objects (helmet lights, SRT kit, bolts).
- **Laser Timeout** — 30s · 60s · 2m · 5m · 15m · 30m.
- **Auto Shutdown** — 5 · 10 · 15 · 30 · 60 min · 2h.
- **Delete Saved Shots** — wipe any pending readings stored in flash.
- **Measure From:** — lets you choose whether to measure from the front or back of the device.
- **Update / Settings** — firmware update and advanced config.
- **Play Snake** — hidden bonus for when you're waiting around.

Advanced: `config.json`

Go to **Menu** → **Update / Settings** → **Edit Settings File**. The DiscoX reboots into USB drive mode and appears as a drive called `DISCOX`. Open `config.json`, edit values, save, eject, then hold the power button to restart. Any setting you omit uses its default.

MEASUREMENT & STABILITY

<code>stability_tolerance</code> default 0.4	Degrees the sensors must converge within before a shot is accepted. Lower = stricter.
<code>stability_buffer_length</code> default 3	Consecutive stable samples required before accepting a reading.
<code>ema_alpha</code> default 0.3	Exponential moving average smoothing. Lower = smoother but slower-reacting.
<code>laser_distance_offset</code> default 0.162	Device length in metres, subtracted from raw laser distance so readings are station-to-station.

LEG DETECTION

leg_angle_tolerance default 1.7	Max difference in degrees between three consecutive shots' bearing/clino to count as the same leg.
leg_distance_tolerance default 0.05	Max difference in metres between three distance readings for leg detection.
laser_wobble default true	Blink the laser on/off in quick succession when a leg is detected.

ANOMALY DETECTION

anomaly_detection default false	Enable warnings when nearby metallic objects may be distorting the magnetic field.
mag_tolerance default 10.0	Magnetometer field-strength tolerance in degrees from expected local field.
grav_tolerance default 10.0	Accelerometer magnitude tolerance in degrees.
dip_tolerance default 10.0	Magnetic dip-angle tolerance in degrees.

CALIBRATION

cal_mag_consistency

default 0.5

Angular consistency window in degrees for the magnetometer during calibration.

cal_grav_consistency

default 0.4

Angular consistency window in degrees for the accelerometer during calibration.

cal_buffer_length

default 5

Samples in the calibration consistency window.

cal_settle_ms

default 250

Milliseconds the device must remain stable before a cal point is accepted.

cal_ema_alpha

default 0.3

EMA smoothing used during calibration.

cal_timeout_ms

default 3000

Max ms to wait for stability before a cal point is force-accepted.

TIMERS & DISPLAY

auto_shutdown_timeout

default 1800

Seconds of inactivity before automatic power-off (1800 = 30 min).

laser_timeout

default 120

Seconds of inactivity before the laser turns off to save power.

screen_brightness

default 255

OLED contrast level, 0–255.

BLUETOOTH

ble_name

default "SAP6_Unicorn"

Bluetooth device name advertised to survey apps. Max 20 characters.